

with an ink outline. This good for simulating the look of cel animation or doing illustrative work. The Ink 'n Paint material has controls for the colour of the paint, as well as controls for the colour and quality of the ink strokes.

### Mental Ray Material

The Mental Ray renderer will render most of 3ds Max's materials, but when it is assigned as the active renderer, a wider array of material types will become available. These include the Mental Ray material along with many other specialty materials.



A 3D scene with multiple mental ray materials applied

## 3.3 Using Maps

Almost any parameter in any of the preceding shaders can be configured with a map. Maps can be created from bitmapped images, movie files, or procedurally. Maps can also contain other maps, creating nested trees of maps. When working with maps, you'll need to navigate up and down through the maps and their component maps. You navigate by using the navigation icons in the Material Editor window.

### Mapping Colour

One of the more common parameters to map is colour. Most shaders use three parameters for colour: Diffuse, Ambient, and Specular.

- The Specular channel is the colour of the object's highlights. Maps on the Specular channel can give the surface a sense of "roughness."
- The Diffuse channel is the "main" colour of the object. This is where bitmaps or procedural textures are added to give the object its character.
- The Ambient channel is the colour of an object in those places where it is in shadow. Typically, this parameter is locked to the Diffuse channel.

### Mapping Opacity

Opacity maps change the transparency of an object. They can be used to create glass- or water-like effects. Varying the transparency by using gradients or falloff maps can give the appearance of varying thicknesses.